

Lab 10: Pareto Fronts

CEVE 421/521

2026-04-10

Draft Material — This content is under development and subject to change.

1 Overview

Monday's lecture used weights to combine objectives — but weights embed value judgments. What if we showed decision-makers the full trade-off instead?

In this lab you'll do two things:

1. **Visualize trade-offs** across 4 objectives using a grid of static house elevations
2. **Use NSGA-II** to find Pareto-optimal adaptive policies that decide when and how much to elevate

This builds on Lab 9's real options idea: instead of committing to a single elevation upfront, an adaptive policy can elevate the house over time in response to observed flood risk. The house elevation model comes from G. G. Garner and K. Keller [1].

Reference:

- G. G. Garner and K. Keller [1] — the adaptive elevation strategy used in this lab

2 Before Lab

Complete these steps BEFORE coming to lab:

1. **Accept the GitHub Classroom assignment** (link on Canvas)
2. **Clone your repository:**

```
git clone https://github.com/CEVE-421-521/lab-10-S26-yourusername.git
cd lab-10-S26-yourusername
```

3. **Open the notebook** in VS Code or Quarto preview
4. **Run the first code cell** — it will install any missing packages (may take 10-15 minutes the first time)

! Important

Submission: You will render this notebook to PDF and submit the PDF to Canvas. See Section 7 for details.

3 Setup

Run the cells in this section to load packages and define the model. You don't need to modify any of this code — just run it and read the comments.

3.1 Packages

```
if !isfile("Manifest.toml")
    import Pkg
    Pkg.instantiate()
end

using SimOptDecisions
using CairoMakie
using DataFrames
using Distributions
using Random
using Statistics

import Metaheuristics # loads the optimization extension

Random.seed!(2026)
```

3.2 Model

The house elevation model simulates a coastal house over a 70-year horizon. Each year, storm surge can flood the house; elevating the house reduces flood damage but costs money upfront.

```
# === Configuration ===

Base.@kwdef struct HouseConfig{T<:AbstractFloat} <: AbstractConfig
    horizon::Int = 70
    house_height::T = 4.0
    house_area_ft2::T = 1500.0
    house_value::T = 200_000.0
end

# === Scenario ===

@scenario def HouseScenario begin
    @timeseries water_levels
```

```

    @continuous dd_threshold
    @continuous dd_saturation
    @continuous discount_rate
end

function sample_scenario(rng::AbstractRNG, horizon::Int)
    # Draw GEV parameters from prior distributions
    μ = rand(rng, Normal(2.8, 0.3))
    σ = rand(rng, truncated(Normal(1.0, 0.15); lower=0.3))
    ξ = rand(rng, truncated(Normal(0.15, 0.05); lower=-0.2, upper=0.5))

    # Generate water level time series from GEV
    surge_dist = GeneralizedExtremeValue(μ, σ, ξ)
    water_levels = [rand(rng, surge_dist) for _ in 1:horizon]

    HouseScenario(;
        water_levels=water_levels,
        dd_threshold=rand(rng, Normal(0.0, 0.25)),
        dd_saturation=rand(rng, Normal(8.0, 0.5)),
        discount_rate=rand(rng, truncated(Normal(0.03, 0.015); lower=0.01, upper=0.07)),
    )
end

# === State and Action ===

struct HouseState{T<:AbstractFloat} <: AbstractState
    elevation_ft::T
end

struct ElevationAction{T<:AbstractFloat} <: AbstractAction
    elevation_ft::T
end

# === Outcome ===

@outcomedef HouseOutcome begin
    @continuous construction_cost
    @continuous npv_damages
end

# === Physics ===

function depth_damage(depth::T, threshold::T, saturation::T) where {T<:AbstractFloat}
    depth <= threshold && return zero(T)
    depth >= saturation && return one(T)
    midpoint = (threshold + saturation) / 2

```

```

    steepness = T(6) / (saturation - threshold)
    return one(T) / (one(T) + exp(-steepness * (depth - midpoint)))
end

function elevation_cost(Δh::Real, area_ft2::Real, house_value::Real)
    Δh <= 0 && return 0.0
    base_cost = 20_745.0
    thresholds = [0.0, 5.0, 8.5, 12.0, 14.0]
    rates = [80.36, 82.5, 86.25, 103.75, 113.75]
    rate = rates[1]
    for i in 1:(length(thresholds) - 1)
        if Δh <= thresholds[i + 1]
            t = (Δh - thresholds[i]) / (thresholds[i + 1] - thresholds[i])
            rate = rates[i] + t * (rates[i + 1] - rates[i])
            break
        end
        rate = rates[i + 1]
    end
    return (base_cost + area_ft2 * rate) / house_value
end
elevation_cost(Δh::Real) = elevation_cost(Δh, 1500.0, 200_000.0)

```

3.3 Policies

We define two policy types.

Static policy: elevate the house once, at the start, to a fixed height.

```

@policydef StaticElevationPolicy begin
    @continuous elevation_ft 0.0 14.0
end

```

Adaptive policy: each year, check if the gap between the floor level and the current water level is less than a `buffer`. If so, elevate by enough to restore the `buffer` plus additional `freeboard`. This is inspired by the adaptive strategy in G. G. Garner and K. Keller [1] — rather than committing to a single elevation upfront, the homeowner responds to observed conditions.

```

@policydef AdaptiveElevationPolicy begin
    @continuous buffer 0.0 6.0
    @continuous freeboard 0.0 6.0
end

```

3.4 Callbacks

These callbacks connect the policies and physics to the simulation framework. You don't need to modify them.

```

SimOptDecisions.initialize(::HouseConfig,      ::HouseScenario,      ::AbstractRNG)      =
HouseState(0.0)
SimOptDecisions.time_axis(config::HouseConfig, ::HouseScenario) = 1:(config.horizon)

# Static policy: elevate at t=1 only
function SimOptDecisions.get_action(
    policy::StaticElevationPolicy,    state::HouseState,    t::TimeStep,
scenario::HouseScenario
)
    if is_first(t)
        return ElevationAction(value(policy.elevation_ft))
    end
    return ElevationAction(0.0)
end

# Adaptive policy: buffer/freeboard rule each year
function SimOptDecisions.get_action(
    policy::AdaptiveElevationPolicy,    state::HouseState,    t::TimeStep,
scenario::HouseScenario
)
    floor_level = 4.0 + state.elevation_ft # house_height + current elevation
    water_level = scenario.water_levels[t]
    gap = floor_level - water_level

    buffer_val = value(policy.buffer)
    freeboard_val = value(policy.freeboard)

    if gap < buffer_val
        needed = (buffer_val - gap) + freeboard_val
        max_additional = 14.0 - state.elevation_ft
        actual = min(needed, max_additional)
        return ElevationAction(max(0.0, actual))
    end
    return ElevationAction(0.0)
end

function SimOptDecisions.run_timestep(
    state::HouseState, action::ElevationAction, t::TimeStep,
    config::HouseConfig, scenario::HouseScenario, rng::AbstractRNG
)
    new_elevation = state.elevation_ft + action.elevation_ft
    construction_cost = elevation_cost(action.elevation_ft, config.house_area_ft2,
config.house_value)

    water_level = scenario.water_levels[t]
    floor_level = config.house_height + new_elevation

```

```

        flood_depth = water_level - floor_level
        damage      = depth_damage(flood_depth,    value(scenario.dd_threshold),
value(scenario.dd_saturation))

        return  (HouseState(new_elevation),  (construction_cost=construction_cost,
damage_fraction=damage))
    end

function SimOptDecisions.compute_outcome(
    step_records::Vector, config::HouseConfig, scenario::HouseScenario
)
    construction_cost = sum(r.construction_cost for r in step_records)
    npv_damages = sum(
        step_records[t].damage_fraction * discount_factor(value(scenario.discount_rate),
t)
        for t in eachindex(step_records)
    )
    return HouseOutcome(; construction_cost, npv_damages)
end

```

i Note

Key design choice: `run_timestep` applies whatever elevation the action specifies — it does NOT gate on `is_first(t)`. For the static policy, `get_action` returns zero after the first timestep, so only one elevation happens. For the adaptive policy, `get_action` can return a nonzero elevation in any year, and `compute_outcome` sums ALL construction costs across the full horizon.

3.5 Metrics

When comparing policies, we don't look at individual scenario outcomes. Instead, we aggregate across all scenarios into summary **metrics**. The optimizer uses these metrics as objectives.

```

"""
Compute metrics from simulation outcomes across many scenarios.
Used by `optimize()` to evaluate candidate policies.
"""

function calculate_metrics(outcomes)
    damages = [value(o.npv_damages) for o in outcomes]
    costs = [value(o.construction_cost) for o in outcomes]
    sorted_damages = sort(damages)
    n = length(damages)

    return (
        construction_cost=mean(costs),
        expected_damage=mean(damages),
    )
end

```

```

    q95_damage=sorted_damages[max(1, ceil(Int, 0.95 * n))],
    cvar_98=mean(sorted_damages[max(1, ceil(Int, 0.98 * n)):end]),
  )
end

```

Four metrics capture different aspects of performance:

- **construction_cost**: average upfront spending across scenarios
- **expected_damage**: average NPV of flood damages
- **q95_damage**: 95th percentile damage (Value at Risk)
- **cvar_98**: average of the worst 2% of damage outcomes (Conditional Value at Risk)

3.6 Scenarios

```

config = HouseConfig()
n_scenarios = 500
rng = Random.Xoshiro(2026)
scenarios = [sample_scenario(rng, config.horizon) for _ in 1:n_scenarios]

println("Config: $(config.horizon)-year horizon, \$(Int(config.house_value)) house")
println("Scenarios: $(n_scenarios) sampled futures")

```

4 Section 1: Brute-Force Trade-offs

Before running any optimizer, let's evaluate every static elevation from 0 to 14 ft (in 0.5 ft steps) across all 500 scenarios. This brute-force grid lets us see the full shape of the trade-off space.

4.1 Run the Grid

```

policies = [StaticElevationPolicy(; elevation_ft=e) for e in 0.0:0.5:14.0]
result = explore(config, scenarios, policies; progress=false)

```

Extract the four metrics for each elevation:

```

elevations = collect(0.0:0.5:14.0)
npv_damages = Array(result[:npv_damages])
construction = Array(result[:construction_cost])

metrics_df = DataFrame(;
  elevation=elevations,
  construction_cost=[mean(construction[p, :]) for p in axes(construction, 1)],
  expected_damage=[mean(npv_damages[p, :]) for p in axes(npv_damages, 1)],
  q95_damage=[
    sort(npv_damages[p, :])[max(1, ceil(Int, 0.95 * n_scenarios))] for
    p in axes(npv_damages, 1)
  ],
  cvar_98=[

```

```

        mean(sort(npv_damages[p, :])[max(1, ceil(Int, 0.98 * n_scenarios)):end]) for
    p in axes(npv_damages, 1)
    ],
)

```

i Note

Checkpoint: At 0 ft elevation, construction cost should be zero and damage metrics should be at their highest. At 14 ft, construction cost should be at its highest and damage metrics near zero. If your results don't follow this pattern, check that the setup cells ran successfully.

4.2 Pair Plot

The pair plot shows every pair of objectives as a scatter plot. Each point is one static elevation, colored by elevation height.

```

let
    obj_names = ["Construction\nCost", "Expected\nDamage", "95th %ile\nDamage", "CVaR
98%\nDamage"]
    obj_cols = [:construction_cost, :expected_damage, :q95_damage, :cvar_98]
    n_obj = length(obj_cols)

    fig = Figure(; size=(800, 800))

    for row in 1:n_obj
        for col in 1:n_obj
            if row == col
                # Diagonal: label
                ax = Axis(fig[row, col]; limits=(0, 1, 0, 1))
                hidedeclarations!(ax)
                hidespines!(ax)
                text!(ax, 0.5, 0.5; text=obj_names[row], align=(:center, :center),
fontsize=14)
            elseif row > col
                # Lower triangle: scatter
                ax = Axis(fig[row, col])
                scatter!(
                    ax,
                    metrics_df[!, obj_cols[col]],
                    metrics_df[!, obj_cols[row]];
                    color=metrics_df.elevation,
                    colormap=:viridis,
                    markersize=8,
                )
            if col == 1

```



```

        ax.ylabel = obj_names[row]
    else
        hideydecorations!(ax; ticks=false, grid=false)
    end
    if row == n_obj
        ax.xlabel = obj_names[col]
    else
        hidexdecorations!(ax; ticks=false, grid=false)
    end
    else
        # Upper triangle: empty
        ax = Axis(fig[row, col])
        hidedecorations!(ax)
        hidespines!(ax)
    end
end
end

    Colorbar(fig[:, n_obj + 1]; colormap=:viridis, limits=(0, 14), label="Elevation
(ft)")
    fig
end

```

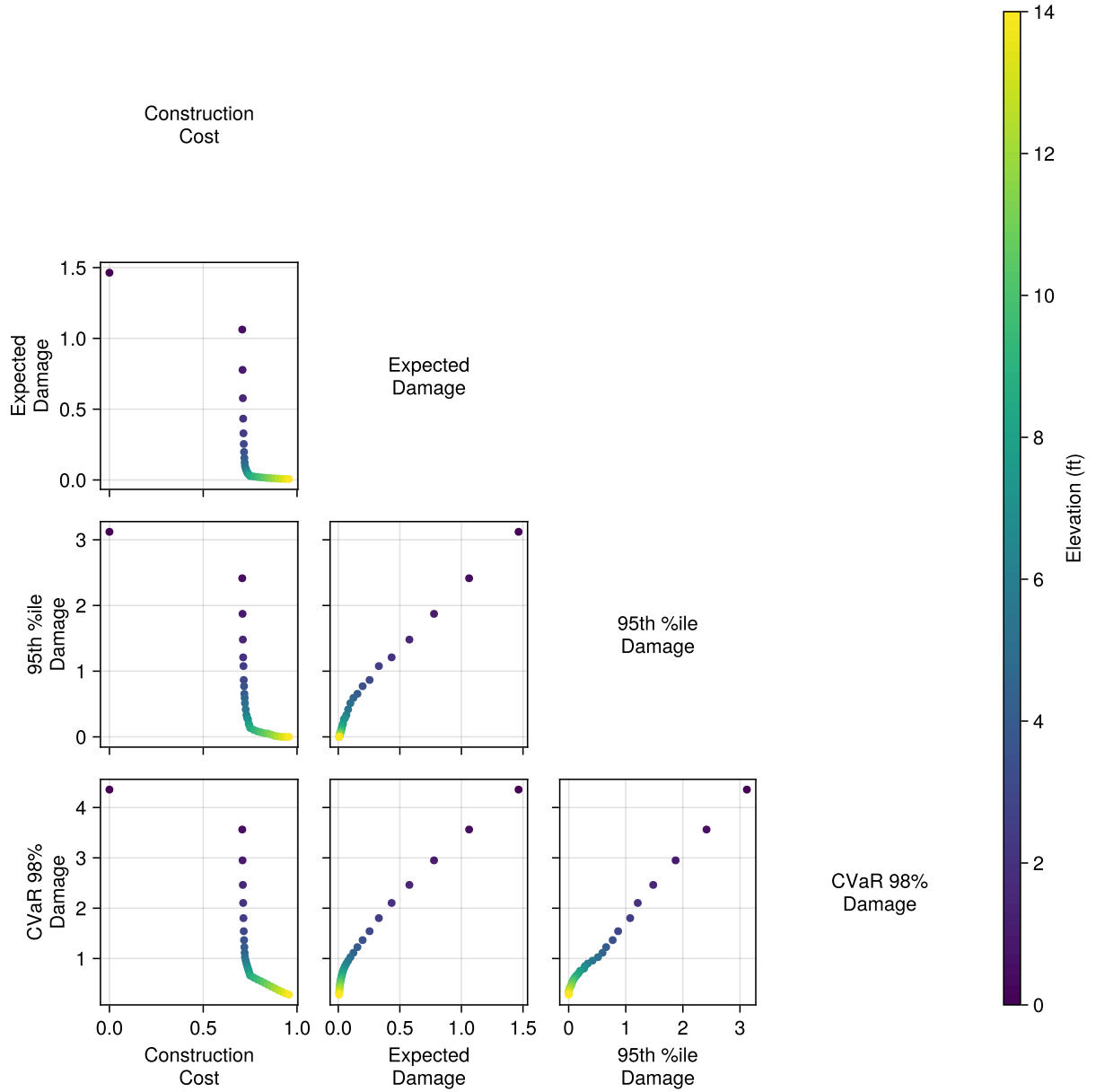


Figure 1: Pair plot of four objectives across static elevations (0–14 ft). Points colored by elevation height. Off-diagonal panels show pairwise scatter; diagonal panels show objective labels.

4.3 Non-Dominated Solutions

A solution is **non-dominated** (Pareto-optimal) if no other solution is better on every objective simultaneously. Let's identify which static elevations are non-dominated across all four metrics.

```
function is_nondominated(i, df, obj_cols)
    for j in axes(df, 1)
        i == j && continue
        all_leq = all(df[j, c] <= df[i, c] for c in obj_cols)
        any_lt = any(df[j, c] < df[i, c] for c in obj_cols)
```

```

        if all_leq && any_lt
            return false
        end
    end
end
return true
end

obj_cols = [:construction_cost, :expected_damage, :q95_damage, :cvar_98]
nondom_mask = [is_nondominated(i, metrics_df, obj_cols) for i in axes(metrics_df, 1)]
nondom_df = metrics_df[nondom_mask, :]
println("$ (nrow(nondom_df)) of $ (nrow(metrics_df)) policies are non-dominated")

```

4.4 Parallel Axis Plot

The parallel axis plot shows each non-dominated solution as a polyline connecting its four objective values. Each axis is normalized to [0, 1] so the objectives are comparable.

```

let
    obj_cols = [:construction_cost, :expected_damage, :q95_damage, :cvar_98]
    obj_labels = ["Construction\nCost", "Expected\nDamage", "95th %ile\nDamage", "CVaR
98%\nDamage"]
    n_obj = length(obj_cols)

    # Normalize against non-dominated solutions only (spreads trade-offs across full
    [0,1] range)
    mins = [minimum(nondom_df[:, c]) for c in obj_cols]
    maxs = [maximum(nondom_df[:, c]) for c in obj_cols]

    fig = Figure(; size=(700, 450))
    ax = Axis(
        fig[1, 1];
        xticks=(1:n_obj, obj_labels),
        ylabel="Normalized value",
        title="Non-Dominated Static Elevations",
        limits=(0.5, n_obj + 0.5, -0.1, 1.1),
    )

    # Draw vertical axis lines
    for j in 1:n_obj
        vlines!(ax, [j]; color=:gray, linewidth=1)
        text!(ax, j, -0.05; text=string(round(mins[j]; digits=3)), align=:center, :top),
        fontsize=9)
        text!(ax, j, 1.05; text=string(round(maxs[j]; digits=3)),
        align=:center, :bottom), fontsize=9)
    end
end

```

```

# Draw polylines for non-dominated solutions
colors = cgrad(:viridis, nrow(nondom_df))
for (idx, row) in enumerate(eachrow(nondom_df))
    vals = [(row[c] - mins[j]) / max(maxs[j] - mins[j], 1e-10) for (j, c)
in enumerate(obj_cols)]
    lines!(ax, 1:n_obj, vals; color=colors[idx], linewidth=2, alpha=0.7)
    scatter!(ax, 1:n_obj, vals; color=colors[idx], markersize=6)
end

Colorbar(
    fig[1, 2];
    colormap=:viridis,
    limits=(minimum(nondom_df.elevation), maximum(nondom_df.elevation)),
    label="Elevation (ft)",
)
fig
end

```

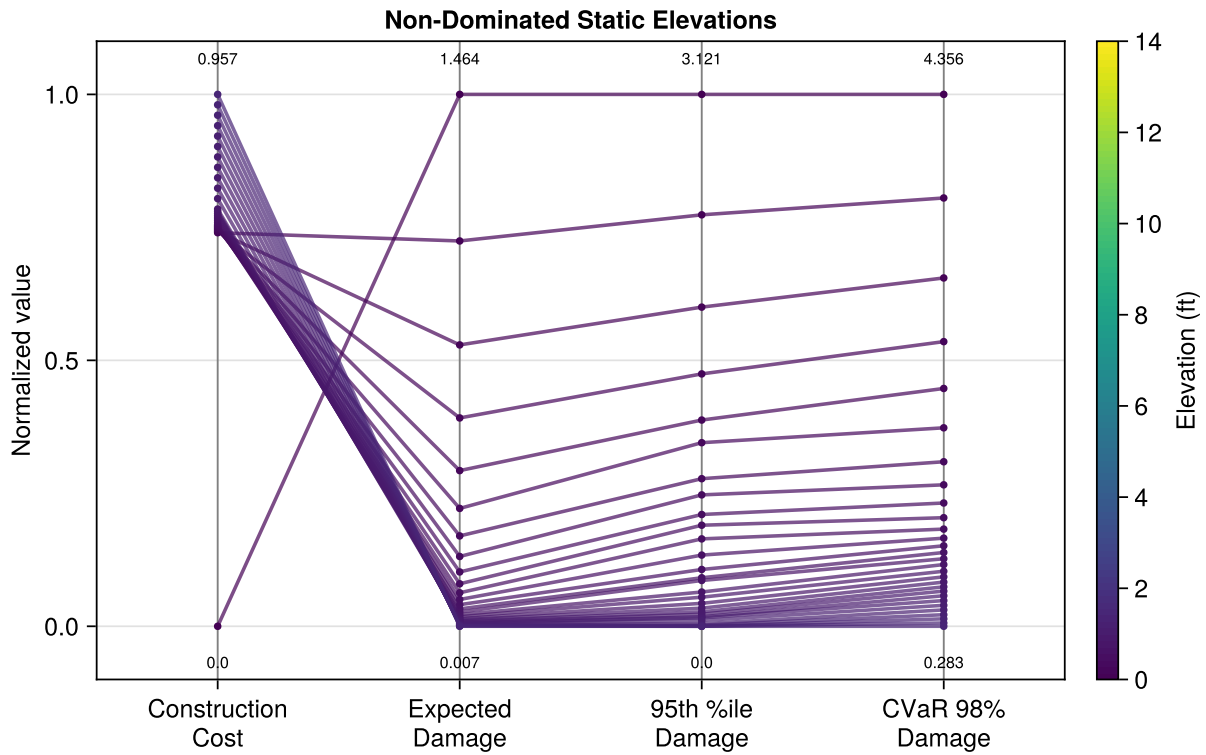


Figure 2: Parallel axis plot of non-dominated static elevations. Each line is one Pareto-optimal policy. Axes normalized to $[0, 1]$ with min/max labels.

Your Response — Question 1

Look at the pair plot and the parallel axis plot.

- 1. Which pairs of objectives are in tension (trade off against each other)? Which move together?*
- 2. Are any of the four objectives redundant — i.e., could you drop one without losing information about the trade-offs?*
- 3. If you had to pick just two objectives to optimize, which would you choose and why?*

5 Section 2: Multi-Objective Optimization

The brute-force grid only works for one-parameter policies (static elevation). The adaptive buffer / freeboard policy has two parameters, making grid search less practical. Instead, we'll use **NSGA-II** — a multi-objective evolutionary algorithm — to search for Pareto-optimal adaptive policies.

This approach mirrors G. G. Garner and K. Keller [1], where direct policy search was used to find adaptive strategies for house elevation under sea-level rise uncertainty.

5.1 Choose Your Objectives

Pick two of the four metrics to optimize. The `minimize(:name)` syntax tells the optimizer which metrics to minimize.

```
# Your code here
```

i Note

Available objectives (all to be minimized):

- `:construction_cost` — average upfront spending
- `:expected_damage` — average NPV of flood damages
- `:q95_damage` — 95th percentile damage
- `:cvar_98` — average of the worst 2% of damage outcomes

5.2 Configure the Optimizer

```
backend = MetaheuristicsBackend(  
    algorithm=:NSGA2,  
    max_iterations=100,  
    population_size=30,  
    parallel=true,  
)
```

5.3 Run Optimization

```
# Your code here
```

i Note

This optimization may take 1–3 minutes depending on your machine. The algorithm evaluates thousands of candidate policies, each simulated across all 500 scenarios.

5.4 Visualize the Pareto Front

Plot the adaptive Pareto front alongside the static grid results for comparison. Use your two chosen objectives as the axes.

Your code here

Figure 3

Your Response — Question 2

Pick one solution from the adaptive Pareto front. What buffer and freeboard values does it use? Describe in plain language what this policy does — when does it trigger elevation, and how much does it elevate by? Who might prefer this solution over the alternatives on the Pareto front?

Your Response — Question 3

Compare the adaptive Pareto front to the static grid points. Does the adaptive policy dominate some or all static solutions on your two chosen objectives? What is the source of the adaptive policy's advantage? Can you think of a situation where a static (one-time) elevation might actually be preferable?

6 Wrap-Up

Key takeaways:

1. **Choosing objectives is a value judgment** — the Pareto front changes depending on which metrics you optimize. The analyst's job is to present trade-offs, not prescribe a single answer.
2. **The Pareto front separates analysis from decision** — it shows what is technically achievable. Stakeholders with different risk tolerances, budgets, and values can each identify their preferred solution.
3. **Adaptive policies can dominate static ones** — by conditioning actions on observed conditions, adaptive strategies exploit information revealed over time, achieving better trade-offs than any fixed design.

7 Submission

1. Write your answers in the response boxes above.
2. **Render to PDF:**
 - In VS Code: Open the command palette (Cmd+Shift+P / Ctrl+Shift+P) -> "Quarto: Render Document" -> select Typst PDF

- Or from the terminal: `quarto render index.qmd --to typst`
3. **Submit the PDF** to the Lab 10 assignment on Canvas.
 4. **Push your code** to GitHub (for backup):

```
git add -A && git commit -m "Lab 10 complete" && git push
```

Checklist

Before submitting:

- ☐ Packages load without error
- ☐ Setup cells run (model, policies, callbacks, metrics, scenarios)
- ☐ Static grid evaluated via `explore()` and metrics extracted
- ☐ Pair plot displayed (Figure 1)
- ☐ Non-dominated solutions identified and parallel axis plot displayed (Figure 2)
- ☐ Question 1 answered (trade-offs between objectives)
- ☐ Two objectives chosen for optimization
- ☐ NSGA-II optimization run and Pareto front found
- ☐ Pareto front plotted with static grid overlay (Figure 3)
- ☐ Question 2 answered (interpret one adaptive solution)
- ☐ Question 3 answered (compare adaptive vs static)
- ☐ Notebook renders to PDF without errors

Bibliography

- [1] G. G. Garner and K. Keller, "Using Direct Policy Search to Identify Robust Strategies in Adapting to Uncertain Sea-Level Rise and Storm Surge," *Environmental Modelling & Software*, vol. 107, pp. 96–104, 2018, doi: 10.1016/j.envsoft.2018.05.006.